



School of Technology Management & Engineering
STUDENT'S WEEKLY DIARY/ WEEKLY LOG

Week	Week 1
Date Range	14th January – 21st January 2026
Time of Arrival	10:00 AM
Time of Departure	6:00 PM
Department/Division	Comp. Biology, Bioinformatics and Crosstalk
Name of the finished task/product	Research orientation, LLM study, AWS reverse proxy setup
Name of the External Mentor (with email id)	pratikchandrani@gmail.com
<i>Description of the main tasks carried out during the week</i>	
- Reviewed the Biomni GitHub repository. It works as an agent framework where an LLM uses pre-defined tools via subprocess execution, with system prompts governing response format and role. - Conducted literature review on reverse proxy solutions: Pangolin, Tailscale, and Headscale. Tailscale suited for VPN, Pangolin for combined reverse proxy and VPN. Concerns noted about Pangolin documentation and stability. - Completed AWS EC2 setup: created IAM roles, SSH key pairs, and security groups. Installed Docker and Pangolin on the instance. Configured domain, Traefik, and Gerbil. - Studied Biomni-R0 in depth: 2-phase training (SFT + RL). Reward design uses ground-truth metric + formatting reward. Each task is constrained to its own tool environment. - Reviewed RL fundamentals: PPO (uses separate reward model), DPO (uses reward function directly), and GRPO (programmatic verification). Also explored LLM steering via activation vector injection. - Studied technical reports for Gemma3, Qwen3, and Kimi K2.	

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Week	Week 2
Date Range	22nd January – 29th January 2026
Time of Arrival	10:00 AM
Time of Departure	6:00 PM
Department/Division	Comp. Biology, Bioinformatics and Crosstalk
Name of the finished task/product	LLM comparison, Biomni testing, Mamba & Nemotron study
Name of the External Mentor (with email id)	pratikchandrani@gmail.com
<i>Description of the main tasks carried out during the week</i>	
• Started and progressed on LLM comparison PPT covering Kimi K2 vs Qwen3 vs Gemma3, focusing on architectural choices, data strategies, and relevance to Biomni. • Completed Qwen3 architecture coverage in PPT: 3 pre-training stages, post-training (Long CoT, Reasoning RL, Thinking Budget, Thinking Fusion), and benchmark evaluations against other models. • Performed test runs of Biomni with gemma3:4b, qwen3:8b, and deepseek-r1:7b. All models struggled with correct code blocks and response formatting. • Drafted a brief implementation plan: SFT using LoRA, RL techniques, LLM steering, guardrails, and optional evaluation of commercial LLMs (ChatGPT, Claude). • Reviewed Kimi K2.5 blog: introduced Agent Swarm framework trained with Parallel-Agent RL (PARL), capable of 100+ sub-agents and 1500+ tool calls. Uses critical steps metric and staged rewards to encourage parallelism. • Studied Mamba architecture in depth: SSMs, discrete SSM, S4, selective SSM/S6, parallel scan, Mamba blocks, and Mamba-2. Mamba offers 5x throughput over Transformers with linear memory scaling. • Started Nemotron-3-Nano report: hybrid Mamba-Transformer MoE (30B-A3B). Uses Phi-4 and Qwen3 for data cleaning/generation. Base model outperforms Qwen3-30B-A3B consistently.	

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Week	Week 3
Date Range	30th January – 6th February 2026
Time of Arrival	10:00 AM
Time of Departure	6:00 PM
Department/Division	Comp. Biology, Bioinformatics and Crosstalk
Name of the finished task/product	Nemotron report, project scoping, dataset planning, fine-tuning tests
Name of the External Mentor (with email id)	pratikchandrani@gmail.com
<i>Description of the main tasks carried out during the week</i>	
• Completed Nemotron-3-Nano report: post-training process, NeMo Gym RL framework (agent/model/resource servers). RLVR outperforms SFT across all models. Pure RL viable for long-context base models. • Finished NVIDIA Nemotron PPT coverage. It is the most promising alternative to Qwen3 due to hybrid Mamba-Transformer design and linear complexity scaling. • Meeting for defining the project scope and finalized Biomni implementation plan. • Presented LLM model selection study PPT. VRAM feasibility analysis performed for 160GB setup. • Tested supervised fine-tuning pipeline on Google Colab using LoRA. Performed dataset conversion to conversational input-output pairs and ran sample benchmark procedure.	

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Week	Week 4
Date Range	9th February – 18th February 2026
Time of Arrival	10:00 AM
Time of Departure	6:00 PM
Department/Division	Comp. Biology, Bioinformatics and Crosstalk
Name of the finished task/product	RL research, DSPy study, Biomni deployment, Open WebUI updates
Name of the External Mentor (with email id)	pratikchandrani@gmail.com
<i>Description of the main tasks carried out during the week</i>	
• Studied production LLM serving frameworks. • Tested LoRA fine-tuning pipeline systematically: rank 32/64, alpha variations, 4-bit vs 16-bit quantization. Model converges around 400 steps (effective batch size 16). Target training loss: 0.25–0.35 for larger models. Recommended starting config: rank 32, alpha 32. • Studied DSPy framework for agentic LLM programming: Predict module, Chain of Thought, structured signatures with Literal types, multi-hop workflows with search. Implemented example modules including structured text extraction and multi-hop claim verification. • Biomni environment on local workstation: fixed Docker routing, resolved “model not found” error. Set up persistent data storage via Docker volume mounts so that user data remains even when rebuilding containers. • Updated Open WebUI from v0.6.34 to v0.6.40. WebSocket error persisted. Disabled RAG pipeline for file uploads since Biomni handles its own processing.	

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